FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness

FlashAttention uses tiling to prevent repeated read and write operations from GPU HBM memory.

By computing attention in blocks loaded into fast SRAM, we reduce memory traffic between HBM and compute units

FlashAttention Implementation Details and Algorithms

Algorithm 1: FlashAttention forward pass with SRAM block sizing and online softmax scaling.

Standard attention materializes the intermediate N x N attention matrix to GPU HBM causing memory bottlenecks.